Rising Creatinine on Trial — TCMR vs ABMR Rejection

1. Rising Cr ≠ automatic rejection

WHAT YOU SEE
Trial banner, creatinine on trial

WHAT IT ENCODES
A rising creatinine in a transplant means graft dysfunction with a broad differential — not automatically rejection. Work up systematically with imaging, Doppler, and biopsy before treating.

BOARD TRAP
Don't reflexively pulse steroids for a creatinine bump — exclude obstruction, dehydration, CNI toxicity, and BK first.

2. ATN — early graft dysfunction (ischemic)

WHAT YOU SEE
Defendant 1: ATN, early post-transplant ischemia

WHAT IT ENCODES
Delayed graft function from acute tubular necrosis (ischemia-reperfusion) is common in the first days, especially deceased-donor kidneys with long cold-ischemia time. Supportive; recovers over time.

BOARD TRAP
Early post-transplant oliguria is often ATN/delayed graft function, not rejection — biopsy if it fails to recover, but don't over-immunosuppress.

3. CNI toxicity — afferent constriction

WHAT YOU SEE
Defendant 2: CNI toxicity, narrowed vessel

WHAT IT ENCODES
Calcineurin-inhibitor nephrotoxicity causes afferent vasoconstriction and rising creatinine that mimics rejection. Check trough level; the treatment is REDUCING the CNI, not intensifying immunosuppression.

BOARD TRAP
CNI toxicity and rejection both raise Cr but need OPPOSITE actions — level and biopsy decide; guessing wrong harms the graft.

4. BK nephropathy — reduce immunosuppression

WHAT YOU SEE
Defendant 3: BK nephropathy, viral cytopathic

WHAT IT ENCODES
BK polyomavirus nephropathy presents as graft dysfunction; urine decoy cells, plasma BK PCR, and SV40+ biopsy confirm. Management is REDUCING immunosuppression — antivirals are not reliably effective.

BOARD TRAP
BK nephropathy mimics rejection histologically — if you raise immunosuppression for presumed rejection you worsen BK; PCR/SV40 stain distinguishes.

5. Recurrent / de novo glomerulonephritis

WHAT YOU SEE
Defendant 4: recurrent GN, blocked tubule

WHAT IT ENCODES
Original glomerular disease (e.g., FSGS, IgA, MPGN, anti-GBM) can recur in the graft, or de novo GN can arise — presenting with proteinuria and rising Cr. Biopsy clarifies.

BOARD TRAP
New proteinuria post-transplant may be recurrent native glomerular disease (classically FSGS), not rejection — biopsy, don't assume.

6. Biopsy — the gold standard

WHAT YOU SEE
Gavel/biopsy, 'don't convict without evidence'

WHAT IT ENCODES
Allograft biopsy is the diagnostic gold standard to separate TCMR, ABMR, CNI toxicity, BK, and recurrent disease, and to grade severity (Banff). Doppler first to exclude vascular/obstructive causes.

BOARD TRAP
You cannot reliably diagnose rejection type clinically — biopsy is required; treating empirically risks the wrong (harmful) therapy.

7. TCMR — T-cell-mediated rejection (tubulitis)

WHAT YOU SEE
TCMR slide: tubulitis, interstitial lymphocytes

WHAT IT ENCODES
T-cell-mediated rejection shows interstitial mononuclear infiltrate, tubulitis, and endothelialitis. First-line treatment is pulse IV steroids; steroid-resistant cases get ATG.

BOARD TRAP
TCMR = cellular; treat with steroids ± ATG. Don't use plasmapheresis/IVIG (those are for antibody-mediated rejection).

8. ABMR — antibody-mediated rejection (C4d+)

WHAT YOU SEE
ABMR slide: capillaritis, C4d staining

WHAT IT ENCODES
Antibody-mediated rejection features peritubular capillaritis, C4d deposition, and donor-specific antibodies (DSA). Treatment: plasmapheresis + IVIG ± rituximab/bortezomib to remove/suppress antibody.

BOARD TRAP
ABMR needs antibody-directed therapy (plasmapheresis/IVIG ± rituximab), NOT just steroids — C4d+ and DSA are the giveaways.

9. C4d + DSA = antibody-mediated

WHAT YOU SEE

C4d / DSA markers on ABMR panel

WHAT IT ENCODES

C4d staining in peritubular capillaries plus circulating donor-specific antibodies confirms ABMR. These markers distinguish it from purely cellular (TCMR) rejection on biopsy.

BOARD TRAP

C4d+ / DSA+ = ABMR (antibody therapy); a C4d-negative tubulitis picture is TCMR (steroids) — the markers drive the treatment choice.

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10. Steroid-resistant rejection → ATG

WHAT YOU SEE

Decision band: steroid-resistant → ATG

WHAT IT ENCODES

Acute cellular rejection not responding to pulse steroids is treated with anti-thymocyte globulin (ATG), a T-cell-depleting agent. This is a defined escalation step.

BOARD TRAP

Steroid-resistant CELLULAR rejection → ATG; don't jump to plasmapheresis (that's for antibody-mediated, not steroid-resistant TCMR).

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11. Doppler ultrasound — first imaging

WHAT YOU SEE

Imaging/ultrasound reference near gavel

WHAT IT ENCODES

Doppler is the first imaging step to exclude obstruction, renal artery/vein stenosis or thrombosis as causes of rising creatinine before biopsy. Cheap, fast, non-invasive.

BOARD TRAP

Image (Doppler) before biopsy to rule out surgical/vascular causes — a rising Cr from RAS or obstruction shouldn't be treated as rejection.

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12. Boards decision band

WHAT YOU SEE

Bottom band: asymptomatic→consider rejection, biopsy, TCMR vs ABMR, BK reduce IS

WHAT IT ENCODES

Algorithm: rising Cr → Doppler then biopsy → TCMR (tubulitis, C4d−) = steroids → ATG if resistant; ABMR (capillaritis, C4d+, DSA) = plasmapheresis + IVIG ± rituximab/bortezomib; BK → reduce immunosuppression.

BOARD TRAP

The cardinal split: TCMR→steroids/ATG (raise/redirect immunosuppression) vs BK→reduce immunosuppression — biopsy + C4d/DSA + BK PCR decide, never treat blind.
